Partial Multiple Imputation With Variational Autoencoders: Tackling Not at Randomness in Healthcare Data

Ricardo Cardoso Pereira, Pedro Henriques Abreu, and Pedro Pereira Rodrigues

***Abstract*—Missing data can pose severe consequences in critical contexts, such as clinical research based on routinely collected healthcare data. This issue is usually handled with imputation strategies, but these tend to produce poor and biased results under the Missing Not At Random (MNAR) mechanism. A recent trend that has been showing promising results for MNAR is the use of generative models, particularly Variational Autoencoders. However, they have a limitation: the imputed values are the result of a single sample, which can be biased. To tackle it, an extension to the Variational Autoencoder that uses a partial multiple imputation procedure is introduced in this work. The proposed method was compared to 8 state-of-the-art imputation strategies, in an experimental setup with 34 datasets from the medical context, injected with the MNAR mechanism (10% to 80% rates). The results were evaluated through the Mean Absolute Error, with the new method being the overall best in 71% of the datasets, significantly outperforming the remaining ones, particularly for high missing rates. Finally, a case study of a classification task with heart failure data was also conducted, where this method induced improvements in 50% of the classifiers.**

***Index Terms*—Healthcare data, missing data, missing not at random, partial multiple imputation, variational autoencoder.**

I. Introduction

MISSING Data (MD) is an issue found in most real-world datasets, and it can be described by the absence of values in one or more features. This issue has a direct impact in the analysis conducted with the datasets, often compromising the success of predictive tasks, statistical studies, and many others [1]. Missing values can have different characteristics and be described by their likelihood of being missing. In fact, three types of MD, called missing mechanisms, are defined [2], [3]: Missing Completely At Random (MCAR), which states there is no relation between the missingness causes and any data; Missing At Random (MAR), which happens when the missing values are related to some part of the observed values (e.g., specific values of a feature); and Missing Not At Random (MNAR), where the reasons behind the missingness cannot be determined from the available data and, therefore, are related to unobserved data. This latter mechanism can have two different settings: the missing values are related to themselves or to other features that are not available in the dataset.

A real-world context that often suffers from MD under the MNAR mechanism is clinical research based on routinely collected healthcare data. The existence of this data relies on the premise that patients receive services from an healthcare system and that physicians detect and report all the relevant information from each clinical appointment. Considering human error and other external factors (e.g., work overload) in this process, it is frequent to have important information not being reported [4]. Besides, there are an unlimited number of confounders that may be unknown or simply unmeasured, and therefore can not be considered by physicians. Other scenarios are studies that require a follow up where physicians contact the patient to evaluate the clinical situation on a given time frame, which often leads to having more missing information since this contact easily fails. Even when it does not fail, the obtained information is sparse compared to the data obtained from hospitalized patients, since physicians have direct access to them [4]. All these scenarios can potentially cause MD under MNAR assumptions, which lead to biased studies.

A common approach to address the MD issue is called imputation, where the missing values are replaced by plausible estimates. Many statistical and machine learning-based methods exist for this purpose, and they tend to produce good results with the MCAR and MAR mechanisms. However, when considering the MNAR mechanism, performing the imputation task is still a challenge since the missing values can not be modeled by the available data. The solutions tend to rely on studying the reasons behind the missingness or on sensitivity analysis, where multiple scenarios are considered to evaluate how resilient are the results [5].

A recent trend in imputation that has been showing good results particularly for MNAR is the use of generative models,

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namely Variational Autoencoders (VAE) [6]–[9]. By generating samples that are based on the training data but diverge towards new concepts, the VAE tends to meet the MNAR assumptions by reducing the bias. However, this method still poses a limitation when dealing with MNAR: it only provides a single imputation result, which may have random bias given the stochastic behavior of the reparameterization trick. To tackle this limitation, we propose in this work an extension to the vanilla (i.e., standard) VAE that uses a partial Multiple Imputation (MI) strategy, which we call Partial Multiple Imputation with Variational Autoencoders (PMIVAE). Several studies show the benefits of applying MI strategies in MNAR scenarios [10], [11], but such strategies were never combined with VAEs. Moreover, considering that the VAE's latent space is a probability distribution, MI can be applied through sampling operations, which would not be possible with other generative models (e.g., Generative Adversarial Networks). Therefore, to the best of the authors' knowledge, this constitutes a novel approach in the MD field.

Our experimental setup considered 34 public datasets from the medical context, 8 state-of-the-art imputation approaches that were compared with the PMIVAE, and 5 missing rates from 10% to 80% achieved by artificially generating the missing values. With this high number of datasets, we aim to cover data scenarios with different characteristics and create an experimental setup that is representative of the MD issue in clinical research. The missing values were artificially generated under the MNAR assumptions [12], [13], and the imputation results were evaluated through the Mean Absolute Error (MAE) between the imputed data and the ground truth. These results were proved to have a statistical significance of 5% through the Three-Way ANOVA on ranks and post-hoc Tukey's HSD tests. Moreover, an analysis of the imputation results on different probability distributions is also presented, followed by an experiment in a multivariate setting with other deep learning-based models and a case study for a heart failure dataset, where the impact of the imputation in a classification task is addressed.

The remainder of the paper is organized in the following way: Section II presents related work that uses VAEs for imputation; Section III describes the PMIVAE method here proposed; Section IV describes the experimental setup and results; Section V extends the experimental setup to encompass a multivariate setting; Section VI presents a case study of the imputation impact on a classification task with heart failure patients; and Section VII presents the conclusions and future directions of this work.

II. Related Work

An Autoencoder (AE) is a type of neural network that tries to learn a compressed representation of the input data with the purpose of outputting a reconstructed version of that same data. It is trained in an unsupervised way, since the labels are not required for the reconstruction purpose. An AE presents an architecture divided into two parts: an encoder, which goes from the input layer to the latent space layer, and a decoder, that goes from the latent space to the output layer. The number of units usually decreases towards the latent space and increases

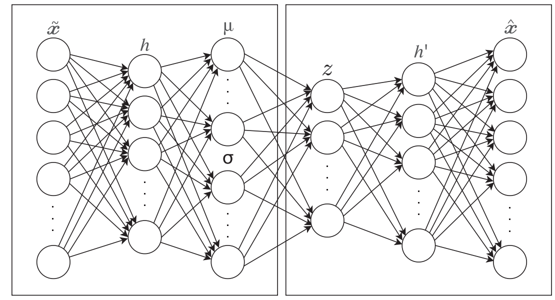

Fig. 1. General architecture of a Variational Autoencoder [6].

again upon the output layer, although other approaches can be followed. The training aims to optimize the model parameters in order to minimize the reconstruction error between the input data and the values obtained from the output layer. The optimization procedure is the same as for any neural network, being usually based on a Stochastic Gradient Descent variant, and using as loss function the Mean Squared Error or the Binary Cross-Entropy (for binary scenarios) [14].

Vanilla AEs are usually used for dimensionality reduction and other compression tasks, but two other variants have been used for MD imputation: the Denoising Autoencoder (DAE), which is a vanilla AE that learns from a corrupted version of the input data, and MD is a type of corruption [15]; and the Variational Autoencoder (VAE), which learns the Gaussian distribution multivariate parameters of the input data, namely the mean and standard deviation, and samples from these to generate new instances with similar characteristics, therefore presenting generative capabilities [16] (see Fig. 1 for a graphical representation of this architecture).

Initially only DAEs were used for the imputation task. Beaulieu-Jones and Moore [17] introduced DAEs to impute missing values in electronic health records, comparing them with Singular Value Decomposition, k-Nearest Neighbors (kNN) imputation, SoftImpute and mean/median imputation. The ALS Pooled Resource Open-Access Clinical Trials (PRO-ACT) dataset was used for the experiments, which encompassed missing rates between 10% and 50%. The DAE achieved the best Root Mean Square Error (RMSE) results for the MNAR mechanism, immediately followed by the kNN and SoftImpute methods. Gondara and Wang [18] compared DAEs with the Multiple Imputation by Chained Equations (MICE), using 14 synthetic and real-world healthcare datasets. The DAEs outperformed MICE for all settings, with improvements above 20% in accuracy and RMSE for the MCAR and MNAR mechanisms. This work was later extended [19] to add multiple imputation to the DAEs, which once again led to better results by this method. More recently VAEs have been used to perform the imputation task, mostly in their vanilla form. McCoy *et al.* [9] compared a VAE with the Principal Component Analysis (PCA) method, used with MI, and also with the mean imputation. The experiments were conducted with a Simulated Milling Circuit dataset, considering 20% and 90% missing rates under MCAR assumptions. The VAE presented the best results, with improvements of 45% over the PCA and 46% over the mean imputation. Boquet *et al.* [7], [20] optimized a 2-layer neural network for regression with data imputed through a VAE, a vanilla Autoencoder and the PCA method. The experimental setup was based on the Caltrans Performance Measurement System (PeMS) dataset, covering 10%, 20% and 40% missing rates under MCAR and MNAR assumptions. The best results were obtained by the network trained with the data imputed by the VAE, with this one showing RMSE improvements of 40% for MNAR and 17% for MCAR. Xie *et al.* [21] used a VAE to perform regression while dealing with the imputation of missing values, but also compared it with other approaches for this latter task: deletion of the missing values, mean imputation and the PCA method. The experiments used industrial data collected from the distributed control system of a polyester plant in China, and considered 10%, 30% and 50% missing rates. The VAE showed the best results for all settings, presenting Mean Square Error (MSE) values below 0.05 for all missing rates, with the worst MSE value being 0.2 for the PCA method.

A few other works have proposed extensions to the VAE. Nazabal *et al.* [22] proposed an extension to handle heterogenous data by modeling different data types with different likelihood models, using an independent neural network for each feature. A hierarchical structure is then applied to handle dependencies between features. Such approach allows VAEs to use other distributions besides the Gaussian, particularly for categorical data. The method was compared with the mean imputation, MICE, a general latent feature model for heterogeneous data and a Generative Adversarial Network for imputation. The experimental setup covered 6 public datasets from the University of California, Irvine (UCI) repository (Adult, Breast, Default-Credit, Letter, Spam and Wine) and missing rates of 10% and 50%. The results are promising for categorical variables, where the extension outperforms the remaining methods in 4 of the 6 datasets used, achieving error decreases over 50%. However, the results for continuous variables were worse, making this extension more suitable for datasets containing mostly categorical features. Fortuin *et al.* [23] trained a VAE with time series data containing missing values, and fed the latent space representations to a Gaussian process that models the time series. The authors state that with this approach the modeling procedure is able to account for correlation between different channels in a multivariate scenario. A logistic regression was then trained with the data imputed through this extension and several other methods: forward and mean imputation, a simple Gaussian process, a vanilla VAE, the VAE extension from [22], and two recurrent neural network-based methods (GRUI-GAN and BRITS). The experiments used real-world medical data from the 2012 Physionet Challenge, under a 60% missing rate. The best AUC-ROC results were obtained when the proposed extension was used, although the improvements were rather small (less than 0.05).

Besides Autoencoders, other deep learning-based methods have been used to perform missing data imputation. Yoon *et al.* [24] adapted the well-known Generative Adversarial Nets to perform imputation, introducing the Generative Adversarial Imputation Nets (GAIN). More recently, Awan *et al.* [25] introduced the Conditional Generative Adversarial Imputation Network (CGAIN), which extends the GAIN method by performing independent imputation on the different data classes, therefore accounting for the individual class distributions. Liu and Zhang [26] proposed a deep learning-based approach composed by two stages: a deep nonparametric reconstruction model is trained with complete data and, afterwards, this model is iteratively used to impute incomplete instances through an optimization process based on coordinate descent. Although most deep learning-based imputation works are targeted for tabular data, time-series data is also addressed. For example, Chan *et al.* [27] incorporated weighted historical data into a neural network approach for traffic prediction and routing.

In conclusion, VAEs have been showing promising results when used to perform the imputation of missing values, although they rely on a single value estimation. Therefore, using VAEs with a MI-based strategy should improve these results, since the random bias from the estimation process is mitigated. This combination was never addressed and is the main novelty of this work. Furthermore, addressing the impact of this imputation approach in healthcare data is a very significant contribution of this work, since this context suffers frequently from MD under MNAR assumptions.

III. Partial Multiple Imputation With Variational Autoencoders

The VAE model has a loss function with two terms: the reconstruction error and a regularizer (see (1), where $q(z|X)$ is the encoder output, $p(X|z)$ is the decoder output, X is the input data and z represents the new samples). The regularizer is the Kullback-Leibler divergence, which forces the disentanglement of the latent space, leading to a well-structured space where similar input data generates similar latent representations. Moreover, it also helps mitigate overfitting [16].

$$L(X) = -E_{z \sim Q(z|X)}[logp(X|z)] + KL(q(z|X) \parallel p(z)) \tag{1}$$

In order for the loss function to be differentiable the sampling process must be stochastic. Therefore, the reparameterization trick is applied. This process adds a random variable, sampled from a normal distribution with $\mu = 0$ and $\sigma = 1$, to the sampling process. This ensures the distribution parameters remain as the learnable parameters of the network [16].

Although the VAE has presented good results in several MD imputation scenarios, it still poses a major limitation: the imputed values are still the result of a single imputation procedure. Given the stochastic nature of the model, caused by the reparameterization trick, the new values being generated can have a random bias. To tackle this issue we propose a partial MI approach, based on a Monte-Carlo sampling, that is applied to the VAE generative phase. We call it Partial Multiple Imputation with Variational Autoencoders (PMIVAE).

The main aspects of the proposed approach are described in Algorithm 1 through pseudocode. To generate a new instance, the VAE takes a sample from the Gaussian parameters previously learned and decodes it. In our approach, the sampling process is repeated N times, leading to N latent space representations

Algorithm 1: Pseudocode for the PMIVAE method.

Input: $train\_rows$, $rows\_to\_impute$, N
Output: $imputed\_rows$
$pre\_train\_rows \leftarrow$ Pre-impute $train\_rows$ with the features' mean
$pre\_rows\_to\_impute \leftarrow$ Pre-impute $rows\_to\_impute$ with the features' mean
$vae\_model \leftarrow$ Train a VAE with pre_train_rows
$enc\_rows\_to\_impute \leftarrow$ Encode $pre\_rows\_to\_impute$ with vae_model
$\mu_{vae}, \sigma_{vae} \leftarrow$ Extract Gaussian parameters from $enc\_rows\_to\_impute$
for each i in $\{1, \ldots, N\}$ **do**
  $x_i = \mu_{vae} + y_i * \sigma_{vae},\ y_i \sim \mathcal{N}(0, 1)$
  Append x_i to X
end for
$X_{mean} \leftarrow$ Average of X
$imputed\_rows \leftarrow$ Decode X_{mean} with vae_model
Return $imputed\_rows$

of new instances. These samples are aggregated into a single one by an averaging operation, and the resulting sample is then decoded. We denote this procedure as a partial MI strategy since for it to be a standard MI the N results from the sampling process need to be independently analyzed before the aggregation, which is not applicable in this context. The goal is to reduce bias and uncertainty in the generative process by accounting for the stochastic behavior of the model with this partial MI strategy. Furthermore, the integration of the MI with the VAE is a key aspect to achieve better results. By applying this procedure directly to the latent space of the VAE model, the multiple samples are generated only from the relevant information learned by it. If the MI was performed in a pre-processing stage, the additional noise in the input data would propagate to the MI results.

The PMIVAE method requires a number of iterations (N), as most MI procedures do. To understand the impact of this parameter in the results, we conducted a sensitivity analysis study that considered 5 values for N (10, 20, 50, 100 and 200), 5 datasets and a missing rate of 40% (which represents a moderate amount of missing values). The remaining settings of the experiment are the same used for the main experimental results of this work (see Section IV). We only considered a subset of the datasets and missing rates used in the main experiments because the time complexity, imposed by testing several N values, was very high. Furthermore, we settled the maximum N value as 200 since it is a reasonable limit when comparing to the iterations used by other works [28]. The study results, evaluated through the Mean Absolute Error (MAE), are presented in Table I. As expected, we see that the best results tend to be achieved with higher N values, although some differences are small. This indicates that the results should be similar with any of the N values. Nevertheless, $N = 200$ iterations still lead to the best results, and therefore we decided to use it for the main experiments.

TABLE I
SENSITIVITY ANALYSIS FOR THE NUMBER OF ITERATIONS. THE BEST MAE RESULTS ARE BOLDED

Dataset	Number of Iterations (N) 10	20	50	100	200
alzheimer-v1	0.1397	0.1402	0.1394	0.1395	**0.1390**
bupa	0.0868	**0.0854**	0.0855	0.0856	**0.0854**
ecoli	0.1127	**0.1125**	0.1126	0.1128	0.1126
pima	0.1072	0.1066	0.1069	0.1065	**0.1064**
wisconsin	0.0598	0.0585	0.0580	0.0578	**0.0577**

IV. Experimental Setup

To evaluate the imputation results of the PMIVAE method, an experiment was conducted where this method was compared with its established state-of-the-art competitors, presented in Table II together with their respective configurations. All Autoencoder-based methods (PMIVAE, VAE and DAE) used a similar neural-network architecture, which is described in Table III. This architecture and the configurations for the remaining methods were defined through a grid search procedure. Furthermore, the Autoencoder-based methods were implemented with the Keras library, while the remaining methods were directly used from the Scikit-learn library. The implementation of PMIVAE method is available on GitHub[1].

The experiment was executed with 34 public datasets from the medical context, which cover different pathologies and domains of clinical research based on routinely collected healthcare data. All datasets are public (most of them can be downloaded from the UCI repository), and they are very heterogeneous in regard to the number of instances and types of features, as Table IV shows: they range from 68 to 2126 instances, from 1 to 19 categorial features, and from 1 to 22 continuous features.

Moreover, the 34 datasets include only complete data: to perform the experiment in a controlled manner, these datasets were injected with missing values under the MNAR mechanism, following the strategy proposed in [13], where the lowest values of the continuous features are removed upon a certain missing rate (see Fig. 2 for a high-level representation of this approach). Each feature was iteratively injected with missing values and imputed, while the imputation results were assessed through the MAE metric applied between the ground truth (i.e., complete dataset) and the imputed data. This strategy was chosen to maximize the coverage of the experiments and to avoid bias by randomly selecting features. Also, this metric was chosen so that all errors had the same weight in the final results. The overall MAE for a dataset is the mean MAE of all its features.

All data was normalized within [0,1] and split in train and test sets following a 70%-30% ratio. For the Autoencoder-based methods, 20% of the training set is used for validation purposes. The categorical features were transformed with the one-hot encoding procedure (i.e., dummy coding). The injection of missing values previously described was performed independently for each of these sets, in order to ensure equal missing rates and MNAR assumptions for all of them. The experiment considered 5 missing rates (10%, 20%, 40%, 60% and 80%), which cover

[1]https://github.com/ricardodcpereira/PMIVAE

TABLE II
BASELINE OF IMPUTATION METHODS USED FOR COMPARISON

Method	Description
Mean Imputation	The imputed value is the mean of the available values for each feature.
Multiple Imputation by Chained Equations (MICE)	Applies a MI strategy in the fitting process of several Bayesian ridge regressions using a round-robin approach with 10 iterations, and assuming as the dependent variables the features containing missing values [29].
k-Nearest Neighbors (kNN) Imputation with $k = 5$	Finds the k nearest neighbors of the instance being imputed using the Euclidean distance between the features without MD, and imputes using the average values from the k instances [30].
SoftImpute	Performs matrix completion through an iterative approach based on nuclear-norm regularization, where the missing values are replaced by estimations from a soft-thresholded SVD [31].
Denoising Autoencoder (DAE)	Vanilla Autoencoder that learns from a noisy variant of the input data. This variant is created through a corruption process, which in this context is the amputation of values [32].
Variational Autoencoder (VAE)	Vanilla VAE (i.e., the basis for the PMIVAE), but used to perform single-sample estimation [9].

TABLE III
NETWORK ARCHITECTURE OF THE AUTOENCODER-BASED METHODS

Aspect	Configuration
DAE Structure	Single hidden layer with 100 units and the ReLU activation function.
VAE Structure	Three layers with 100 units: two "parallel" layers (mean and variance), followed by the layer that samples points from the latent Gaussian distribution. This is the basic configuration for any VAE.
Optimization Algorithm	Adam, with the Mean Squared Error as the loss function and a learning rate of 0.001.
Training Procedure	Batches of 64 instances, a maximum of 200 epochs, an early stopping rule that stops training if the validation loss as no improvements over 100 epochs, and a reduction of the learning rate by 80% using again the previous criteria.
Overfitting Mitigation	Each layer uses L2 regularization with a 0.01 factor.
Network Initialization	Network weights are randomly initialized by sampling from a truncated normal distribution centered on zero.
Output	The output layers use the sigmoid activation function, since the data is normalized between $[0, 1]$.

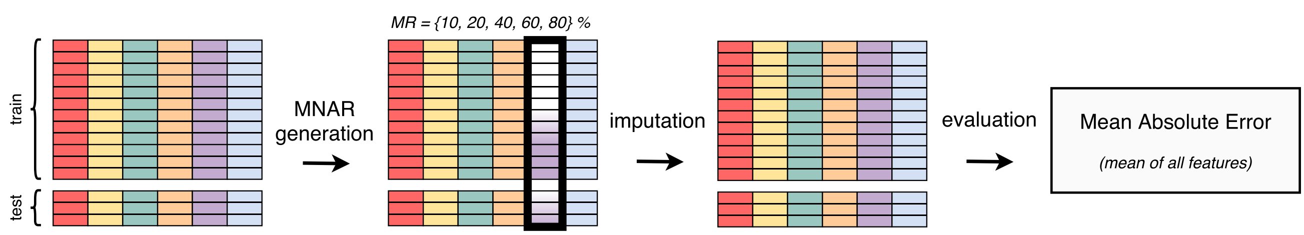

Fig. 2. High-level representation of the experimental setup.

different magnitudes of the MD issue. For the Autoencoder-based methods the missing values were pre-imputed with the mean of the respective features. To mitigate bias and stochastic behaviors, the entire pipeline was executed in 30 independent runs, with the datasets being shuffled in each one. The results considered for comparison are the mean of these runs. Each run was executed in a computer running Windows 10, which had the following specifications: CPU AMD Ryzen 5600X, 16 GB RAM, and GPU NVIDIA GeForce GTX 1060 6 GB. The deep learning-based methods took roughly the same time to train in all settings (around four minutes per dataset), while the remaining methods took between one and five seconds to be executed.

The obtained results were analyzed from 3 different perspectives. The main analysis is a global perception of the imputation error provided by all methods, achieved by comparing the MAE results and validating their statistical significance. To complement this analysis, the previous results were also grouped by the number of datasets where each method outperformed the remaining. Finally, a study to identify the most suitable method for different distributions was conducted. These 3 different analyses are presented in the following sub-sections (IV-A, IV-B and IV-C).

A. Gross Imputation Error Comparison

Regarding the MAE produced by the imputation methods, the average results for all datasets are presented in Table V. Fig. 3 displays the density plots of these results for each imputation method, considering the individual values obtained from each of the 30 runs, which allows for a more comprehensive analysis of the variance among all methods. In an overall analysis, the PMIVAE method surpassed the remaining ones, achieving an average MAE of 0.142, being followed by the VAE and SoftImpute methods with an average MAE of 0.161 and 0.175, respectively. When analyzing the results by missing rate, the PMIVAE method shows the lowest MAE values for the 20%, 40%, 60% and 80%

TABLE IV
CHARACTERISTICS OF THE PUBLIC MEDICAL DATASETS

Dataset	# Instances	# Features	
		Categorical	Continuous
acute-inflammations-nephritis	120	6	1
acute-inflammations-urinary	120	6	1
alzheimer-v1	317	3	7
autism-adolescent	98	19	1
autism-adult	701	16	1
autism-child	288	16	1
bc-coimbra	116	1	9
biomed	194	1	5
breast-tissue-2c	106	1	9
bupa	345	2	5
cleveland_0_vs_4	173	1	13
cmc	1473	8	2
cryotherapy	90	3	4
ctg-2c	2126	1	21
diabetic-retinopathy	1151	4	16
ecoli	336	1	7
fertility-diagnosis	100	8	2
haberman	306	1	3
heart-statlog	270	7	7
immunotherapy	90	3	5
kala-azar	68	2	5
lymphography-v1	142	16	3
new-thyroid-N-vs-HH	215	1	5
newthyroid-v1	185	1	5
parkinson	195	1	22
pima	768	1	8
postoperative-SvsA	86	8	1
relax	182	1	12
saheart	462	2	8
thoracic	470	14	3
thyroid_3_vs_2	703	1	21
transfusion	748	1	4
vertebral-2c	310	1	6
wisconsin	683	1	9

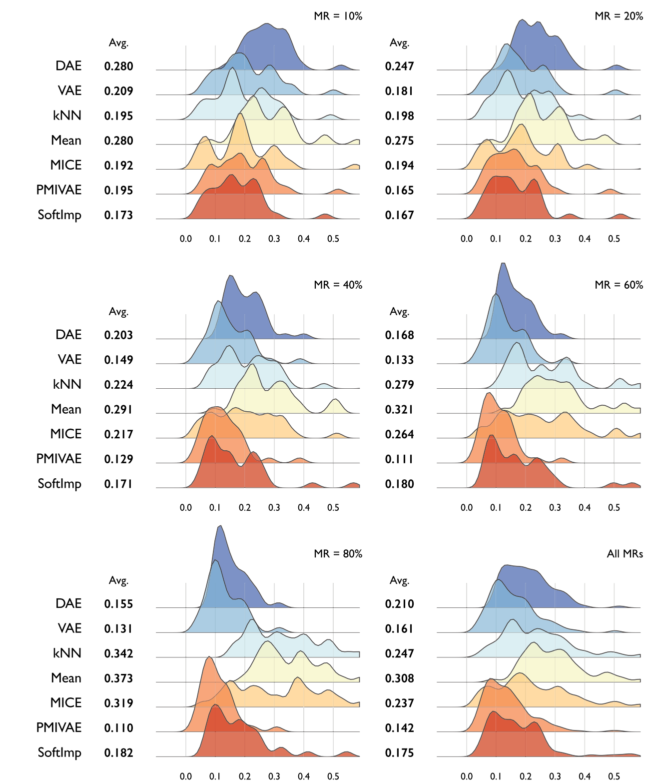

Fig. 3. Density plots of the MAE results obtained for each imputation method. The analysis is presented by missing rate, with the overall results being displayed in the last plot. The average MAE for each setting is also displayed.

missing rates. For the 10% rate it is surpassed by the SoftImpute and MICE methods.

To understand how these results change according to the size of the datasets, Table VI presents the Pearson correlation coefficients calculated between the datasets' size and the respective MAE values for the imputation methods. In this context, the size of the dataset is its total number of values ($size = \#instances * \#features$). All coefficients are negative, which indicates that the imputation results tend to be worse when the size of the dataset increases. However, the coefficients are all within $[-0.5, 0]$, therefore representing low correlation and similar consistence among the different imputation methods.

The benefits of using the PMIVAE method in MNAR scenarios with moderate and high missing rates are related to two main factors: the generative capabilities of the VAE, which lead to imputed values that have similar characteristics to the available data but are diverging towards new concepts, and the reduced level of uncertainty achieved through the partial MI procedure, which allows higher stability in the imputed values. Furthermore, the PMIVAE is surpassed in the 10% missing rate because the amount of information still available in the features is enough for a method more suitable for MAR scenarios (such as the SoftImpute or MICE) to still produce good results. In other words, the MNAR assumptions are weak in this low missing rate scenario, and therefore the level of uncertainty surrounding the missing values is lower since the features probably contain enough information to ease the imputation procedure.

An interesting aspect to highlight is that all Autoencoder-based methods reduce their MAE values when the missing rate increases, while the remaining methods present the opposite behavior. This a clear indicator that Autoencoders are suitable for MNAR scenarios, particularly with high missing rates, since they deal well with corrupted data, mostly because the additional noise leads to less overfitting and improved generalization [33].

To validate the statistical significance of the obtained results, the Three-Way ANOVA on ranks test was applied with a significance level of 5%. The factors considered were the dataset, the missing rate and the imputation method, while the dependent variable was the MAE. The standard Three-Way ANOVA test can only be applied if the normality assumptions are met. In this study such assumptions were violated, and therefore the data was transformed into rankings using the Ordered Quantile normalization [34], which always produces normally distributed transformed data. The same assumptions were ensured for the data subgroups. From the obtained p-values we can conclude the results are statistically significant for all factors: the dataset and imputation method factors show $p = 2e^{-16}$, while the missing rate shows $p = 3.33e^{-06}$.

Finally, to conclude if the PMIVAE outperformed the remaining methods with a statistical significance of 5%, the post-hoc Tukey's HSD test was applied for the imputation method factor. The PMIVAE significantly outperformed all the remaining methods, showing $p < 0.001$ for all settings.

TABLE V
MAE RESULTS OBTAINED FOR EACH IMPUTATION METHOD AND GROUPED BY MISSING RATE. THE REPORTED VALUES ARE THE AVERAGES AND STANDARD DEVIATIONS OF ALL DATASETS. THE BEST RESULTS FOR EACH MISSING RATE ARE BOLDED

Imputation Method	Missing Rate 10%	20%	40%	60%	80%
DAE	0.280 ± 0.08	0.247 ± 0.07	0.203 ± 0.06	0.168 ± 0.05	0.155 ± 0.05
VAE	0.209 ± 0.10	0.181 ± 0.09	0.149 ± 0.07	0.133 ± 0.06	0.131 ± 0.06
kNN	0.195 ± 0.10	0.198 ± 0.11	0.224 ± 0.12	0.279 ± 0.14	0.342 ± 0.13
Mean	0.280 ± 0.10	0.275 ± 0.11	0.291 ± 0.12	0.321 ± 0.12	0.373 ± 0.12
MICE	0.192 ± 0.11	0.194 ± 0.12	0.217 ± 0.14	0.264 ± 0.15	0.319 ± 0.16
PMIVAE	0.195 ± 0.10	$\mathbf{0.165 \pm 0.09}$	$\mathbf{0.129 \pm 0.07}$	$\mathbf{0.111 \pm 0.06}$	$\mathbf{0.110 \pm 0.06}$
SoftImp	$\mathbf{0.173 \pm 0.09}$	0.167 ± 0.10	0.171 ± 0.11	0.180 ± 0.11	0.182 ± 0.10

TABLE VI
PEARSON CORRELATION COEFFICIENTS CALCULATED BETWEEN THE SIZE OF THE DATASETS AND THE RESPECTIVE MAE VALUES FOR THE IMPUTATION METHODS. THE COEFFICIENTS ARE INDEPENDENTLY PRESENTED FOR EACH MISSING RATE

Imputation Method	Missing Rate 10%	20%	40%	60%	80%
DAE	-0.39	-0.34	-0.32	-0.33	-0.31
VAE	-0.45	-0.42	-0.43	-0.45	-0.46
kNN	-0.31	-0.27	-0.26	-0.26	-0.26
Mean	-0.30	-0.27	-0.25	-0.24	-0.25
MICE	-0.32	-0.30	-0.28	-0.31	-0.30
PMIVAE	-0.40	-0.38	-0.35	-0.34	-0.35
SoftImp	-0.29	-0.25	-0.19	-0.15	-0.11

TABLE VII
PERCENTAGE OF DATASETS WHERE EACH IMPUTATION METHOD WAS THE BEST. THE DATA IS GROUPED BY MISSING RATE, WITH THE OVERALL RESULTS BEING PRESENTED IN THE LAST COLUMN. THE BEST RESULTS FOR EACH MISSING RATE ARE BOLDED

Imputation Method	Missing Rate 10%	20%	40%	60%	80%	Overall Balanced
DAE	0%	3%	3%	6%	3%	3%
VAE	0%	0%	0%	0%	0%	0%
kNN	9%	6%	0%	0%	0%	3%
Mean	0%	0%	0%	0%	0%	0%
MICE	18%	15%	6%	3%	0%	8%
PMIVAE	26%	**56%**	**88%**	**88%**	**97%**	**71%**
SoftImp	**47%**	21%	3%	3%	0%	15%

B. Best Imputation Method Per Dataset

Table VII presentes an alternative analysis where the results are grouped by the amount of datasets where each imputation method obtained the lowest MAE. We can conclude that PMIVAE was the overall best method in 71% of the datasets, followed by the SoftImpute method in 15% of them. When analyzing the results by missing rate, we see that PMIVAE also presents the best results, with an exception for the 10% missing rate where the SoftImpute surpassed it (the reasons behind this behavior are the same already presented for the MAE results analysis). Moreover, we see clear improvements in the PMIVAE results when the missing rates increase: for a 20% missing rate it is better in 56% of the datasets, for the 40% and 60% rates it is better in 88% of them, and finally for a 80% rate PMIVAE is better in 97% of the datasets.

C. Sensitivity to Different Data Probability Distributions

Considering that our experimental setup addressed iteratively each feature of the 34 datasets, we decided to perform an analysis focused on the most suitable imputation methods for each Probability Density Function (PDF) available in the data. The goal was to understand if some imputation methods were more suitable for specific distributions. We developed an automated approach to find the PDF of each feature in each dataset, where the most well-known PDFs[2] were iteratively fitted to the data, and the resulting fitting error was calculated through the residual sum of squares between the distribution and the histogram of the original data. The PDF with the lowest error for each feature was chosen as the one that better describes it. Finally, we crossed this information with the best imputation method for each feature, leading to the results available in Fig. 4. This heat-map presents the best imputation method for each PDF identified in the process previously described. To ensure the results were trustworthy, only distributions followed by at least 4 features were considered. The analysis was performed individually for each missing rate, and each cell contains the percentage of features that lead each method to be the best.

The obtained results showed a clear dominance of the PMIVAE method in all PDFs when the missing rate is 40% or higher. For the smaller missing rates (10% and 20%) the variance between the best methods is high, which can be explained by a smaller impact of the imputation procedure: the amount of values generated is not enough to have an impact on the distribution. Nevertheless, for moderate and high missing rates, PMIVAE shows stable results for all PDFs, therefore being a versatile method that can be applied on data with different characteristics.

V. MULTIVARIATE EXPERIMENTAL SETUP

Based on the encouraging results from the main experimental setup, we decided to extend these experiments to encompass a multivariate missing data setting. Furthermore, we also decided to compare the best method from the main experiments (PMIVAE) with two other state-of-the-art generative deep learning-based models: the Missing Data Importance-Weighted Autoencoder (MIWAE) [35] and the GAIN [24] methods. Both models were parametrized according to the best hyperparameters reported by their authors. In this multivariate scenario, the missing data generation follows the same MNAR pattern from the main experiments, but is now applied to the entire dataset. Therefore, the missing rate is global for the dataset, and several features are injected with missing values simultaneously. Such multivariate setting usually leads to increased imputation errors, and to understand if the models present large deviations from the ground truth in specific features we also

[2]See https://docs.scipy.org for the list of PDFs considered.

TABLE VIII

MAE AND RMSE RESULTS OBTAINED FOR EACH DEEP-LEARNING IMPUTATION METHOD AND GROUPED BY MISSING RATE. THE REPORTED VALUES ARE THE AVERAGES AND STANDARD DEVIATIONS OF ALL DATASETS. THE BEST RESULTS FOR EACH MISSING RATE ARE BOLDED

Metric	Imputation Method	Missing Rate 10%	20%	40%	60%	80%
MAE	GAIN	0.317 ± 0.11	0.329 ± 0.11	0.389 ± 0.14	0.627 ± 0.34	$\mathbf{0.800 \pm 0.48}$
	MIWAE	0.409 ± 0.17	0.424 ± 0.18	0.492 ± 0.21	0.748 ± 0.42	0.959 ± 0.61
	PMIVAE	$\mathbf{0.285 \pm 0.10}$	$\mathbf{0.293 \pm 0.11}$	$\mathbf{0.354 \pm 0.15}$	$\mathbf{0.599 \pm 0.36}$	$\mathbf{0.800 \pm 0.55}$
RMSE	GAIN	0.453 ± 0.12	0.473 ± 0.13	0.558 ± 0.16	0.993 ± 0.62	$\mathbf{1.273 \pm 0.81}$
	MIWAE	0.560 ± 0.15	0.580 ± 0.17	0.673 ± 0.22	1.124 ± 0.67	1.472 ± 1.00
	PMIVAE	$\mathbf{0.424 \pm 0.11}$	$\mathbf{0.437 \pm 0.13}$	$\mathbf{0.523 \pm 0.18}$	$\mathbf{0.970 \pm 0.64}$	1.312 ± 0.98

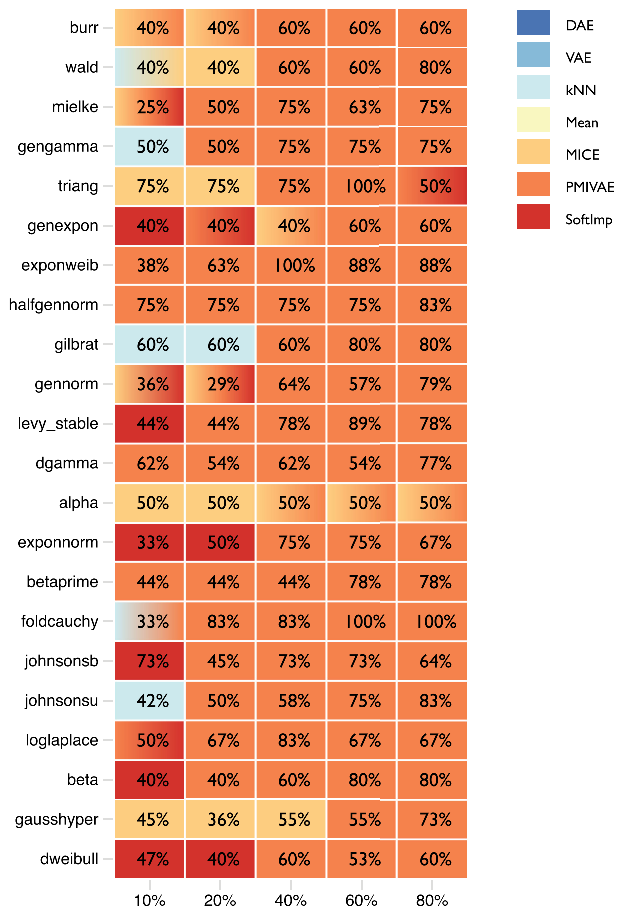

Fig. 4. Heat-map of the best imputation methods for each PDF followed by, at least, 4 features of the 34 datasets. The results are displayed independently for each missing rate (x-axis), and the color legend is displayed in the top-right corner. When 2 methods are tied for the same PDF, a color gradient of both is used. The percentage of features that lead a method to be the best is displayed in each cell.

calculated the Root Mean Squared Error (RMSE) metric. For the remaining aspects, these experiments followed the same configurations of the main experimental setup. The obtained results are presented in Table VIII. The PMIVAE method surpassed the remaining two deep learning models in both MAE and RMSE metrics. PMIVAE was the best method in 75% of the settings combining the five levels of missingness and the 34 datasets. Such results show that the PMIVAE scales well for multivariate scenarios, while keeping the best imputation quality.

VI. A Case Study on Heart Failure

Encouraged by the positive results obtained from the experiments of Sections IV and V, we aimed to assess the impact of the imputation performed by the PMIVAE method in a classification problem. We focused this new experiment in a heart failure dataset available for medical research on the PhysioNet repository since late 2020 [36], [37]. This dataset was constructed with electronic health data collected from 2008 patients admitted with heart failure to a hospital in Sichuan, China, between December 2016 and June 2019. It contained 167 features (125 continuous and 42 categorical) covering regular clinical characteristics (e.g., body temperature, pulse, respiration rate, systolic blood pressure, diastolic blood pressure, mean arterial blood pressure, weight, height, body mass index, type of heart failure, New York Heart Association cardiac function, Killip Grade and Glasgow Coma Scale) and several features extracted from echocardiograms performed on patients (e.g., left ventricular ejection fraction, left ventricular end diastolic diameter, mitral valve peak E and A wave velocities, tricuspid valve regurgitation velocity, and tricuspid valve regurgitation pressure). In a pre-processing stage, 3 features containing unique and constant values were dropped, since they were irrelevant for the classification task. A procedure to remove outliers from the continuous features was also applied, where the values below and above 3 standard deviations from the mean were removed.

The considered classification target was a binary label indicating if the patient would be readmitted to the hospital within 6 months after discharge. Heart failure is a frequent reason for hospitalization of elderly people that leads to a high mortality [36]. Therefore, having a decision support system that can help physicians know in advance what patients are more likely to be readmitted allows a better management of patients and discharges by the hospital systems.

Initially all the dataset instances contained missing values, with the missing rate range varying from 1% to 97.9% in different features. Although there is no clear way of identifying the missing mechanism behind MD, we performed a few statistical and exploratory steps to find the more likely mechanisms. We started by applying the Little's test [38], which gave us a clear indication that the missing values were not under MCAR assumptions. We then performed exploratory data analysis, where we took into account meta-data from the features containing MD and correlations with other features. After aggregating all the information, we concluded that most missing features are likely to be under MNAR assumptions.

TABLE IX
CLASSIFIERS USED IN THE STUDY AND THEIR CONFIGURATIONS

Classifier	Configuration
Artificial Neural Network (ANN)	The network had 2 hidden layers (with 100 and 50 nodes, respectively), ReLU as the activation function, Adam as the optimizer, 2000 epochs and 20% of the training samples for validation.
k-Nearest Neighbors (kNN)	$k = 5$ neighbors were considered.
Random Forest (RF)	100 trees were used.
Support Vector Machine (SVM)	RBF kernel and $\gamma = \frac{1}{N * \sigma^2}$ were used, being N the number of features.

TABLE X
F1 SCORE RESULTS OBTAINED FOR EACH IMPUTATION METHOD APPLIED TO EACH CLASSIFIER. THE REPORTED VALUES ARE THE AVERAGES AND STANDARD DEVIATIONS OF 30 RUNS. THE BEST RESULTS FOR EACH CLASSIFIER ARE BOLDED

Imputation Method	Classifiers' F1 Score			
	ANN	kNN	RF	SVM
DAE	0.538 ± 0.02	0.476 ± 0.03	0.531 ± 0.02	$\mathbf{0.538 \pm 0.02}$
VAE	0.517 ± 0.03	0.471 ± 0.03	0.531 ± 0.02	0.523 ± 0.03
kNN	0.529 ± 0.02	$\mathbf{0.481 \pm 0.02}$	0.530 ± 0.02	0.534 ± 0.03
Mean/Mode	0.528 ± 0.02	0.480 ± 0.03	0.517 ± 0.02	0.530 ± 0.02
MICE	0.532 ± 0.03	0.478 ± 0.02	0.529 ± 0.02	0.531 ± 0.02
PMIVAE	$\mathbf{0.539 \pm 0.03}$	$\mathbf{0.481 \pm 0.02}$	$\mathbf{0.534 \pm 0.02}$	0.535 ± 0.03
SoftImp	0.523 ± 0.03	0.473 ± 0.03	0.532 ± 0.02	0.523 ± 0.02

In order to train the Autoencoder-based methods, a complete set of instances is required. Since all instances of the dataset had missing values, pre-imputation was needed. As in the previous experiments, the features' mean was used for this purpose.

To understand the impact on different types of classification algorithms, the 4 classifiers presented in Table IX were considered in the study [39].

The remaining parameters used for the 4 classifiers are the recommended defaults by the Scikit-learn library. The dataset was divided into training and test sets in a stratified manner (70%-30% ratio), ensuring a balanced split between the label values in the two sets. Before training the classifiers, the SMOTE [40] over-sampling method was applied to solve the imbalance between labels on the training set (the global imbalance ratio between readmitted and other patients was 63%). To evaluate the classification results the F1 Score metric was considered. The same imputation methods used in the general experiments from Section IV were used here. The same parametrization was applied, with a single exception: given the higher dimensionality of this heart failure dataset, the Autoencoder-based methods used two hidden layers (with 90 and 60 units), and a latent space with 30 units. The maximum number of epochs was also extended to 2000.

The experiment was executed 30 independent times, with the dataset being shuffled in each one. The obtained results, which are the mean of these 30 runs, are available in Table X. This classification problem is very complex, leading in general to modest F1 Scores (peaking at 0.539 for the ANN). Although the variance of results between the different imputation methods is small for all classifiers, we see improvements in 50% of them when PMIVAE is used (ANN and RF). For the kNN classifier, the best result is shared between the kNN imputation and PMIVAE. Only the SVM presented the best results with another imputation method (DAE). Therefore, in conclusion, we see general improvements when PMIVAE is used for the imputation task previous to the classification.

VII. CONCLUSION

In this work a new method to perform MD imputation, called Partial Multiple Imputation with Variational Autoencoders (PMIVAE), is introduced. PMIVAE is an extension to vanilla VAEs, and it applies a partial MI strategy to reduce bias and uncertainty in the new values generated. The method was compared with 8 state-of-the-art imputation approaches, in a very complete experimental setup that considered 34 public datasets from the medical context, MD generation under MNAR assumptions and 5 missing rates between 10% and 80%. The results showed that the PMIVAE was the best overall method in 71% of datasets. Moreover, PMIVAE tends to perform better for moderate and high missing rates, being only surpassed in the 10% rate by the SoftImpute method. These results were validated with a statistical significance of 5%. The PMIVAE also surpassed the MIWAE and GAIN methods in a multivariate setting, achieving the best results for most missing rates and datasets. Furthermore, a case study of a classification task with a heart failure dataset was also performed. The PMIVAE method provided improvements in 50% of the considered classifiers.

In the future we want to address the limitation of only considering a single amputation procedure by testing the PMIVAE method with other strategies to generate missing values under MNAR assumptions. Furthermore, we want to extend PMIVAE by incorporating information from other datasets of the same context, which would allow adjustments to the distribution parameters of the VAE that could reduce bias in MNAR settings.

APPENDIX
ACRONYMS

A list of acronyms from this article is available in Table XI.

TABLE XI
ACRONYMS

Definition	Abbreviation
Autoencoder	AE
Denoising Autoencoder	DAE
Generative Adversarial Imputation Nets	GAIN
k-Nearest Neighbors	kNN
Mean Absolute Error	MAE
Mean Square Error	MSE
Missing At Random	MAR
Missing Completely At Random	MCAR
Missing Data	MD
Missing Data Importance-Weighted Autoencoder	MIWAE
Missing Not At Random	MNAR
Multiple Imputation by Chained Equations	MICE
Multiple Imputation	MI
Partial Multiple Imputation with Variational Autoencoders	PMIVAE
Performance Measurement System	PeMS
Pooled Resource Open-Access Clinical Trials	PRO-ACT
Principal Component Analysis	PCA
Probability Density Function	PDF
Random Forest	RF
Root Mean Square Error	RMSE
Support Vector Machine	SVM
University of California, Irvine	UCI
Variational Autoencoder	VAE

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